

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
# =====
# KONFIGURASI HALAMAN (SAMA SEPERTI FILE ASLI)
# =====
st.set_page_config(page_title="Dashboard Analisis Siswa", layout="wide")
st.title("📊 Dashboard Analisis Hasil Simulasi Siswa")
st.markdown("Aplikasi ini digunakan untuk menganalisis hasil simulasi 50 siswa pada 20 indikator penilaian menggunakan pendekatan
```

```
2026-02-27 09:58:06.357 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.359 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.360 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.361 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.362 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.363 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:06.364 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()
```

```
# =====
# INPUT DATA
# =====
uploaded_file = st.file_uploader("Upload file Excel data siswa", type=["/content/data_simulasi_50_siswa_20_soal.xlsx"])

if uploaded_file is None:
    st.info("Silakan upload data terlebih dahulu untuk memulai analisis.")
    st.stop()
```

```
2026-02-27 09:58:44.190 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.193 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.194 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.195 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.196 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.197 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.198 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.199 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.200 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 09:58:44.200 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
```

```
# =====
# MEMBACA DATA
# =====
# The `uploaded_file` variable is None because the Streamlit file uploader
# is not active or a file has not been uploaded.
# To proceed in a Colab environment without a live Streamlit app,
# we can directly specify the file path. It is assumed the file is
# located at "/content/data_simulasi_50_siswa_20_soal.xlsx"
# as indicated by the 'type' parameter in st.file_uploader in the previous cell.
file_path = "/content/data_simulasi_50_siswa_20_soal.xlsx"
df = pd.read_excel(file_path)
```

```
# memastikan numerik
indikator = df.apply(pd.to_numeric, errors="coerce")

st.subheader("📄 Preview Data")
st.write("Menampilkan 5 baris pertama dataset:")
st.dataframe(indikator.head())

st.divider()
```

```
2026-02-27 10:00:05.038 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.040 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.041 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.043 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.044 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.045 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.091 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.092 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:00:05.094 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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2026-02-27 10:00:05.098 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()

```
# =====
# STATISTIK DESKRIPTIF
# =====
st.header("1 Statistik Deskriptif")

rata_soal = indikator.mean()
std_soal = indikator.std()
rata_total = rata_soal.mean()

indeks_kepuasan = (rata_total / 4) * 100

def kategori(x):
    if x >= 80: return "Sangat Baik"
    elif x >= 65: return "Baik"
    elif x >= 50: return "Cukup"
    else: return "Kurang"

col1, col2, col3, col4 = st.columns(4)
col1.metric("Rata-rata keseluruhan", f"{rata_total:.2f}")
col2.metric("Standar deviasi rata-rata", f"{std_soal.mean():.2f}")
col3.metric("Indeks capaian (%)", f"{indeks_kepuasan:.2f}")
col4.metric("Kategori capaian", kategori(indeks_kepuasan))

st.write("Tabel statistik tiap indikator:")
st.dataframe(pd.DataFrame({
    "Rata-rata": rata_soal,
    "Standar deviasi": std_soal
}))

st.divider()
```

2026-02-27 10:01:20.395 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2026-02-27 10:01:20.398 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2026-02-27 10:01:20.399 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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 2026-02-27 10:01:20.435 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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 2026-02-27 10:01:20.444 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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 2026-02-27 10:01:20.447 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2026-02-27 10:01:20.448 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2026-02-27 10:01:20.450 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 2026-02-27 10:01:20.451 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
 DeltaGenerator()

```
# =====
# VISUALISASI DISTRIBUSI NILAI
# =====
st.header("2 Distribusi Nilai Siswa")

fig_dist, ax_dist = plt.subplots(figsize=(8,4))
indikator.mean(axis=1).hist(ax=ax_dist, bins=10)
ax_dist.set_title("Distribusi rata-rata nilai siswa")
ax_dist.set_xlabel("Skor rata-rata")
ax_dist.set_ylabel("Frekuensi")
st.pyplot(fig_dist)

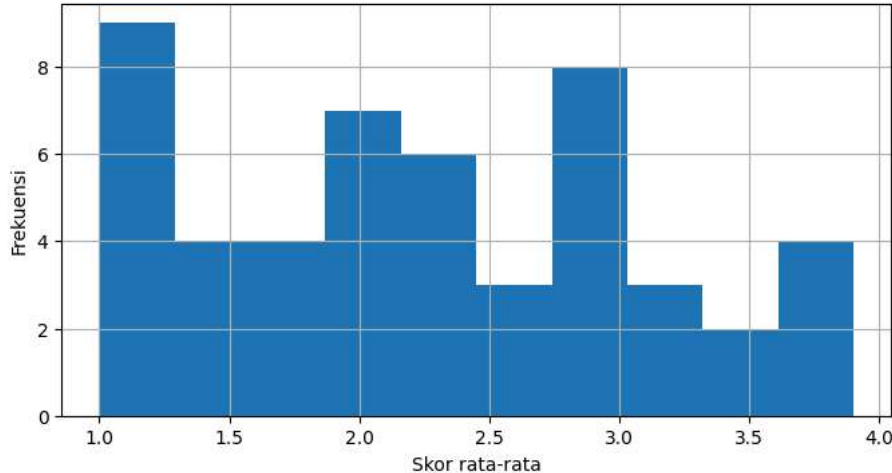
st.divider()
```

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2026-02-27 10:01:48.016 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.021 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.121 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.292 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.293 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.294 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.297 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.299 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:01:48.302 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()

```

Distribusi rata-rata nilai siswa



```

# =====
# ANALISIS GAP
# =====
st.header("📊 Analisis GAP Kinerja")

gap = 4 - rata_soal
soal_prioritas = gap.idxmax()

fig_gap, ax_gap = plt.subplots(figsize=(10,4))
gap.plot(kind="bar", color="orange", ax=ax_gap)
ax_gap.set_ylabel("GAP terhadap skor maksimum")
ax_gap.set_title("Analisis kesenjangan tiap indikator")
plt.xticks(rotation=45)
st.pyplot(fig_gap)

st.warning(f"Indikator prioritas perbaikan: **{soal_prioritas}**")

st.divider()

```

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2026-02-27 10:02:16.708 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:16.709 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:16.710 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:16.768 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.139 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.140 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.140 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.143 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.145 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.147 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.149 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.151 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:17.152 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()

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```

# =====
# KORELASI
# =====
st.header("📈 Korelasi Antar Indikator")

corr = indikator.corr()
fig_corr, ax_corr = plt.subplots(figsize=(9,7))
im = ax_corr.imshow(corr, cmap="coolwarm", vmin=-1, vmax=1)
plt.colorbar(im, ax=ax_corr)

```

```
ax_corr.set_xticks(range(len(corr.columns)))
ax_corr.set_yticks(range(len(corr.columns)))
ax_corr.set_xticklabels(corr.columns, rotation=90)
ax_corr.set_yticklabels(corr.columns)
ax_corr.set_title("Heatmap Korelasi Pearson")
```

```
st.pyplot(fig_corr)
```

```
st.divider()
```

```
2026-02-27 10:02:37.455 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:37.456 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:37.457 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:37.536 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.221 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.222 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.223 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.224 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.226 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:38.227 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()
```

```
# =====
# REGRESI LINEAR
# =====
st.header("5 Analisis Regresi Linear")
```

```
X = sm.add_constant(indikator.iloc[:, :-1])
y = indikator.iloc[:, -1]
```

```
model = sm.OLS(y, X, missing="drop").fit()
```

```
st.write(f"Nilai R2 model: **{model.rsquared:.3f}**")
```

```
koef = model.params[1:].sort_values(ascending=False)
```

```
st.subheader("5 Koefisien terbesar")
st.table(koef.head(5).to_frame("Koefisien"))
```

```
st.divider()
```

```
2026-02-27 10:02:55.693 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.698 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.704 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.730 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.731 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.736 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.745 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.751 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.754 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.760 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.764 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.767 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.768 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.772 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.773 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:02:55.774 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()
```

```
# =====
# SEGMENTASI SISWA
# =====
st.header("6 Segmentasi Siswa (Clustering)")
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(indikator.fillna(0))
```

```
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
cluster = kmeans.fit_predict(X_scaled)
```

```
indikator_cluster = indikator.copy()
indikator_cluster["Cluster"] = cluster
```

```
cluster_mean = indikator_cluster.groupby("Cluster").mean()
ranking = cluster_mean.mean(axis=1).sort_values(ascending=False).index
```

```
mapping = {
    ranking[0]: "Kelompok Tinggi",
```

```

    ranking[1]: "Kelompok Sedang",
    ranking[2]: "Kelompok Rendah"
}

indikator_cluster["Kategori"] = indikator_cluster["Cluster"].map(mapping)

st.write("Contoh hasil segmentasi:")
st.dataframe(indikator_cluster[["Kategori"]].join(indikator).head(10))

st.divider()

```

```

2026-02-27 10:03:31.334 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.336 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.337 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.367 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.368 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.369 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.379 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.381 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.381 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.382 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:03:31.384 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()

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# =====
# INTERPRETASI OTOMATIS
# =====
st.header("📊 Interpretasi Hasil")

st.write(f"""
Berdasarkan analisis, diperoleh rata-rata capaian siswa sebesar **{rata_total:.2f}**
dengan indeks capaian **{indeks_kepuasan:.2f}%** yang termasuk kategori **{kategori(indeks_kepuasan)}**.

Analisis GAP menunjukkan bahwa indikator **{soal_prioritas}**
merupakan aspek yang paling membutuhkan peningkatan.

Hasil regresi menunjukkan bahwa beberapa indikator memiliki pengaruh signifikan
terhadap capaian akhir siswa, sehingga peningkatan pada indikator tersebut
dapat berdampak langsung terhadap hasil belajar.

Segmentasi siswa menunjukkan adanya kelompok kemampuan berbeda,
yang mengindikasikan perlunya strategi pembelajaran yang lebih adaptif.
""")

st.success("Dashboard selesai dianalisis dan siap digunakan untuk laporan penelitian.")

```

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2026-02-27 10:04:52.782 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:04:52.784 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:04:52.786 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:04:52.787 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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2026-02-27 10:04:52.789 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:04:52.791 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-02-27 10:04:52.792 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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DeltaGenerator()

```